

Memory as a Learned Policy:

Budgeted Write Operations for Long-Horizon LLM Agents

Parth Kocheta
University of Maryland

April 28, 2026

Abstract

Long-running language-model agents accumulate context faster than any context window can hold, so they store some of it externally. The question is what to keep. Today’s memory systems—Mem0, Letta, Mastra, Zep, our own KGmem—answer that question with hand-written code. The store grows from use; the control layer that picks what to commit typically does not.

We frame memory as a budgeted decision problem. Given a hard retention budget K , the write policy decides which K items from a conversation to commit to a knowledge graph, and is rewarded if a frozen reader can answer downstream questions from those K items. The reward signal is *reader-overlap*: for each question, take what a frozen reader retrieves from the full conversation text—the set of dialogue turns the reader believes are relevant—and score how many of those turns the post-write knowledge graph actually preserved. No per-question evidence annotation is required; in controlled studies this signal correlates with gold evidence coverage at within-turn Spearman $\rho \approx 0.61$, while gold coverage itself correlates almost perfectly ($\rho \approx 0.98$) with the trivial number-of-turns-kept feature, making it a misleading reward in isolation.

We train a LoRA adapter on a 4B-parameter open-source model with GRPO, where each episode is one conversation turn, the write policy emits up to four tool-call operations into a typed knowledge graph, and the reward blends reader-overlap ($w = 0.7$) with the reader’s LLM-judge-scored answer accuracy on a held-out battery ($w = 0.3$). The reader is held fixed: a hand-coded composite of BM25 over entity text, dense embedding similarity, and named-entity lookup, fused via reciprocal-rank fusion (RRF). An audit on a paired 50-question LoCoMo sample showed this hybrid retriever recovers roughly 75% of the accuracy gap that pure entity-only retrieval introduces, making it strong enough that gains from the learned write policy are not artefacts of an underpowered read backend.

The success criterion is strict. A learned write policy must beat *random subsampling at matched retention budget* on LoCoMo and LongMemEval-S; this is the content-agnostic efficiency frontier any contribution has to clear. It must also beat a BM25-based heuristic write policy that selects turns by query overlap. [TBD: Report headline numbers once the long Phase B run lands. Anchor: random subsampling at $F = 0.1$ on LoCoMo achieves $\sim 4\%$ judge-graded QA, BM25 heuristic at the same F achieves $\sim 63\%$ on LongMemEval-S; the learned policy needs to land above the latter.]

The framing we want to leave the reader with: facts live in the store, strategy lives in the weights. A single trained adapter can run against any user’s local store, with no per-user fine-tuning and no need to move user data into a shared corpus.

1 Introduction

Long-running AI assistants need long-running memory. ChatGPT remembers user preferences across sessions; Claude keeps notes within a project; a coding agent should be able to recall the

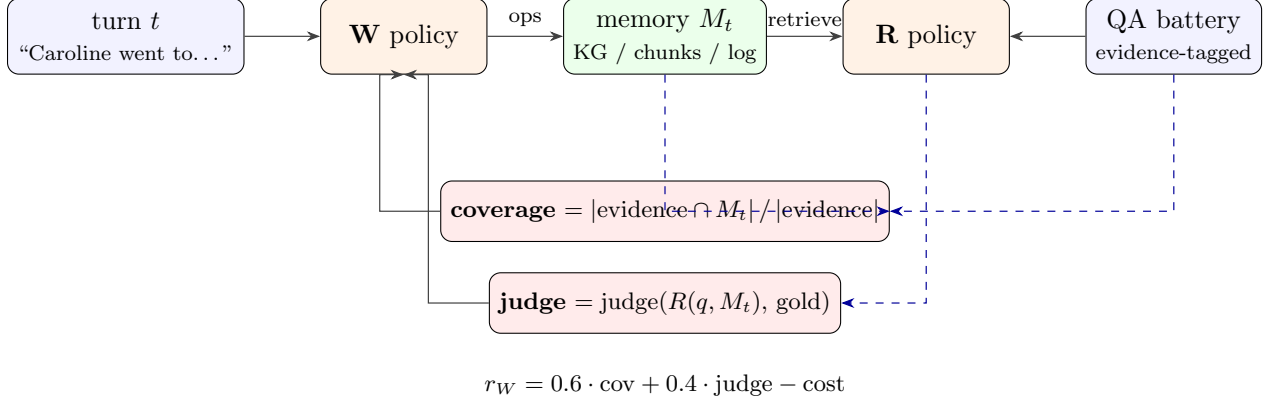


Figure 1: The write policy W ingests one conversation turn and emits memory ops; the read policy R later retrieves from the resulting store M_t to answer questions. W 's reward blends a deterministic *evidence-coverage* term against LoCoMo's per-question dia_id labels with an LLM-judge term over R 's answer. Coverage is dense and free; judge keeps the policy honest about whether the stored content is actually answerable.

design choices it made yesterday. Each of these uses an external memory store paired with a control layer that decides what to write down and what to read back. The store grows from use, but the control layer typically does not.

Where the design choices currently live. Across popular open-source memory systems for LLM agents, a pattern emerges: each pairs an external store with a hand-tuned control layer.

- **Mem0** [3] stores facts as flat key-value pairs. On each new turn, an LLM-as-judge prompt selects one of ADD/UPDATE/DELETE/NONE. The prompt and its thresholds are fixed.
- **Letta / MemGPT** [12] models memory as OS-style tiers (working, recall, archival) and moves items between tiers via prompted function calls. The promotion rules are written by hand.
- **Zep / Graphiti** [14] maintains a temporal knowledge graph in which edges carry valid-time intervals. The rules for when an edge becomes invalid are fixed.
- **Mastra** [9] compresses raw turns into a dated bullet observation log using two background agents (*Observer* and *Reflector*). The compression schedule and prompts are fixed.
- **KGmem** (the lead author's prior system, used as one of our backends) keeps typed entities with state transitions, and resolves new mentions against existing entities through a three-tier string/embedding/LLM cascade. The cascade thresholds (0.85 for string match, 0.85 for embedding accept, 0.65 for reject) are constants chosen by inspection.

None of this is a criticism: hand-tuned control layers are a reasonable starting point and most of these systems work well in practice. The open question we take up is whether the same control decisions can be improved by RL against the same answer-correctness signal already used to evaluate the system, and whether the resulting policy generalises across storage substrates.

What we do. We treat memory operations as actions of a learned policy implemented as a low-rank adapter (LoRA) [5] on a 4B-parameter open-source language model. The actions are tool calls

into a memory backend. We define a single op vocabulary covering both sides of the system: read ops (`memory_search`, `memory_expand`, `memory_filter`, `memory_rerank`) and write ops (`lookup_entity`, `create_entity`, `update_state`, `merge_entities`, `add_relation`, `mark_contradiction`, `forget`, `noop`).

The *read* policy answers questions. Its reward is judged answer correctness, minus a small per-tool-call cost. We train it with GRPO [16], DeepSeek’s critic-free simplification of PPO that has become the default RL algorithm for tool-use training.

The *write* policy ingests a conversation turn at a time. Its reward is deferred: after the write trajectory ends, the read policy answers a held-out battery of questions that depend on the turn the write policy just saw. The reward blends evidence-coverage (a deterministic, dense signal computed from the resulting graph’s provenance) with the read policy’s judged answer accuracy on the battery, minus a storage cost.

The two policies are trained in alternation: train the read policy against the current memory state, then freeze the read policy and train the write policy against the read policy’s evaluation. Repeat.

Why this matters. For research, the contribution is a *learned operating system* for memory—facts in the store, strategy in the weights—and a backend-agnostic op vocabulary that makes the strategy transferable. For deployment, the same observation implies a useful asymmetry: a single trained adapter can run against any user’s personal store, with no per-user fine-tuning needed and no need to move user data into a shared corpus.

Contributions.

1. A reframing of LLM memory as a *budgeted decision problem*: under a hard retention budget K , the learnable layer is the write policy that picks which K items to commit. Read becomes a hand-coded hybrid retrieval stack that the write policy optimises against.
2. A label-free dense reward, *reader-overlap*, that requires no per-question evidence annotations and is robust to the n_{kept} confound that defeats raw evidence-coverage. We show coverage by itself correlates with the trivial turns-kept feature at $\rho \approx 0.98$, making the budgeted formulation necessary.
3. A backend-retriever-aligned evaluation protocol that controls for the dominant confound in memory-system evaluation: an audit on a paired LoCoMo sample shows mismatched retrievers can swing QA accuracy by 0.40 in absolute terms (McNemar $p = 1.9 \times 10^{-6}$). Our read-side retriever fuses BM25, dense, and NER via RRF, and we run training and evaluation on the same retriever.
4. Two clean baselines the learned policy must beat: random subsampling at matched F (the efficiency frontier; trivial to compute, hard to dominate at low F) and a BM25-based heuristic write policy. We report the area under the efficiency curve (AUEC) as the headline metric, with statistical significance via paired McNemar tests stratified by question category.
5. Empirical results on LoCoMo and LongMemEval-S, plus a head-to-head against KGmem’s hand-coded extraction pipeline on personal ChatGPT-export data with self-generated questions. [TBD: Numbers fill in once the long Phase B run completes.]

2 Related Work

2.1 Memory systems for LLM agents

The space splits along two axes: where facts live (in the model weights or in an external store), and how the operations on those facts are controlled (hand-coded or learned).

Facts in an external store, hand-coded ops. Most production systems sit here. **Mem0** [3] stores facts in a vector index and uses an LLM-as-judge prompt for each new turn to choose between ADD, UPDATE, DELETE, or NONE. **Letta** (formerly MemGPT) [12] keeps a tiered memory (working, recall, archival) and uses LLM function calls to move things between tiers. **Zep** / **Graphiti** [14] stores a bi-temporal knowledge graph—each edge has a valid-time interval and an ingestion-time. **Mastra Observational Memory** [9] uses two background agents (Observer and Reflector) to compress the chat history into a dense bullet-point observation log; it reports the highest published score on LongMemEval (94.87% with gpt-5-mini). **Generative Agents** [13] run periodic reflection cycles that condense raw observations into higher-level beliefs. In all of these, the schedule, thresholds, and prompts that decide what to remember are designed by hand.

Facts in the weights. A different line modifies model parameters per user or per fact. **ROME** / **MEMIT** [10] do surgical edits to feed-forward layer activations to insert specific facts. Per-user fine-tuning embeds a user’s history into a LoRA adapter. These approaches give very tight retrieval (no search needed—the model just knows) but fragment the model fleet (one adapter per user) and risk catastrophic forgetting.

Facts in the store, ops learned. This cell is the smallest and is where we sit. The closest precedent is **ACE** [1] (under review at ICLR 2026), which evolves text-form “playbooks” based on agent execution feedback—but the playbook is a single prompt document, not a structured memory store. **Voyager** [18] maintains a Minecraft skill library that grows over time, with the LLM deciding what to add—but additions are trial-and-error, not RL-trained.

2.2 Reinforcement learning for tool use

The general framework is: an LLM emits tool calls as actions, an environment runs them and returns observations, the episode ends with a verifiable reward, and we train the LLM to emit better tool sequences.

RLHF [11] pioneered using RL on language model outputs, but with reward models trained from human preferences and PPO as the algorithm. **PPO** [15] requires a critic network to estimate baseline rewards.

GRPO [16] drops the critic. Instead, for each prompt, sample a group of G rollouts, compute the reward of each, and use the group mean as the baseline. The advantage of rollout i is $(r_i - \bar{r})/\text{std}(r)$. This works exceptionally well when the reward is verifiable (e.g. correct answer to a math problem). DeepSeek-R1 demonstrated GRPO on chain-of-thought; it is now the default RL algorithm for tool-use training.

Search-R1 [6] applies GRPO to multi-turn search-then-answer over Wikipedia. Each rollout is a sequence of search queries followed by a final answer; reported gains are +41% over RAG baselines on NaturalQuestions with Qwen2.5-7B. We use the Tinker-cookbook reproduction of Search-R1 as the structural template for our environment; the present work extends that template to the memory setting and adds a write side.

Reflexion [17] and verbal RL approaches update natural-language “self-reflections” rather than weights. They are fast and cheap but plateau quickly because the policy is a static prompt.

Curriculum learning [2] is a classic finding we make use of: starting an RL agent on easy tasks gives reward signal that bootstraps learning on harder tasks. We use LoCoMo’s question categories as a curriculum (easy single-hop $\rightarrow$ hard multi-hop and adversarial).

2.3 Joint training of two policies

CoSearch [1] (concurrent ICLR work) jointly trains a reasoning policy (an LLM that decides what to search) and a document ranker (a scoring head that re-orders candidates) via RL on agentic search. Their ranker is a learned scoring function, not an action policy in the same sense as our W and R. The shared idea worth taking is *joint training of two model components that interact through a shared substrate*; we extend that pattern to two action policies that share an op vocabulary across read and write sides of memory. **HippoRAG** [4] uses Personalised PageRank over a graph of memories—inspired by the hippocampus—but uses no RL. Our approach extends the joint-training idea from query-time (reasoning+ranking) to ingestion-time (write+read).

2.4 Adapter-based fine-tuning

LoRA [5] adds small low-rank matrices alongside the attention layers of a frozen base model. Fine-tuning only the adapter keeps the base intact and produces a ~ 50 –200 MB artefact that can be applied to any copy of the same base. We use rank-32 adapters on Qwen3-4B-Instruct [7], trained via Tinker—Thinking Machines Lab’s managed RL training service that exposes the right primitives (forward-backward, sample, optim-step) for GRPO.

2.5 What is missing

Despite a crowded space, no published system combines all three:

1. Discrete, interpretable memory operations (Mem0, Letta, and KGmem all have these).
2. Operations chosen by a learned policy supervised by outcome reward (Search-R1 has this for retrieval only).
3. A write side trained against deferred QA accuracy on a held-out battery (no published system).

That is the gap we close.

3 Method

3.1 The op vocabulary

We define a unified action space $\mathcal{A} = \mathcal{A}_R \cup \mathcal{A}_W$. Each action is a JSON tool call with a fixed schema.

Read ops $\mathcal{A}_R$.

- `memory_search(query, k, source)`: Retrieve top- k chunks by lexical (BM25), semantic (dense), or hybrid (RRF [16]) match. The retrieval substrate is whichever Backend the env was built with.

- `memory_expand(seed_uids, k_per)`: Pull adjacent chunks of seeds (one-hop graph follow / sibling-turn lookup).
- `memory_filter(predicate, value)`: Restrict the current hit set by session, speaker, or time window.
- `memory_rerank(strategy, query)`: Re-order hits by dense similarity, recency, or session order.

The episode ends when the model emits a final answer formatted as “**Answer:** ...”.

Write ops $\mathcal{A}_W$.

- `noop`: This turn is not memory-worthy. The default.
- `lookup_entity(query, type, top_k)`: Find existing entities by name. The policy uses this to avoid duplicates.
- `lookup_relation(source_uid, target_uid)`: Inspect existing graph edges.
- `create_entity(name, type, state)`: Add a new entity. Type is one of nine: `person`, `project`, `tool`, `organization`, `belief`, `decision`, `concept`, `period`, `event`, `goal`.
- `update_state(uid, new_state, transition_type, trigger_summary)`: Mutate an entity. Transition type is one of `update`, `contradiction`, `resolution`, `archival`.
- `merge_entities(canonical_uid, alias_uid)`: Collapse two entities, moving transitions and edges.
- `add_relation(source_uid, target_uid, rel_type, description)`: Add a typed graph edge.
- `mark_contradiction(uid, contradicting_state)`: Flag that the current turn conflicts with the entity’s prior state, but do not pick a winner.
- `forget(uid, reason)`: Soft-delete (archive). Keeps the transition history but marks the entity inactive.

3.2 Background: KGmem, a typed-transition knowledge graph

One of the three backends we evaluate predates this paper. We refer to it throughout as *KGmem*; the open-source implementation is the PIE memory system from prior work by the lead author, retained in our repository under `pie/`. We use a renamed reference here for readability and to keep the role of the data structure separate from the role of any specific control layer.

KGmem represents the user’s world as a graph of typed entities (one of `person`, `project`, `tool`, `organization`, `belief`, `decision`, `concept`, `period`, `event`, or `goal`). Each entity carries a history of typed state transitions: every transition records a (*from state*, *to state*) pair labelled with one of `creation`, `update`, `contradiction`, `resolution`, or `archival`. In particular, contradiction is a first-class transition rather than an overwrite—an unusual choice among open knowledge-graph memory systems—which lets both the conflicting and prior states coexist until a later transition resolves them. KGmem also supports per-entity importance scoring and typed relationship edges; this paper uses the entity-plus-transition substrate only.

KGmem matters here for two reasons. First, the typed transitions give the write policy a richer action vocabulary than “store” and “delete”—it can emit `contradiction` when a turn conflicts with an existing entity, leaving both states retrievable for the read policy. Second, entity uids let the read policy chain queries (`lookup_entity` $\rightarrow$ `transitions_for(uid)`) when a question requires

multi-hop reasoning over a single entity’s evolving state. We re-use the data structure; the hand-coded extraction prompts and three-tier resolver that ship with KGmem are replaced by the learned write policy.

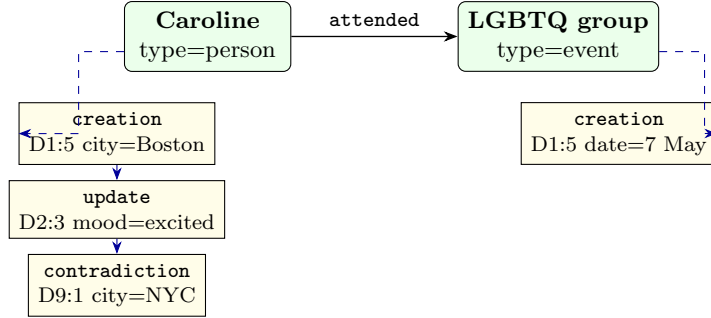


Figure 2: KGmem schema. Solid arrow: typed relationship between entities. Dashed arrows: typed state transitions, each tagged with the dia_id that caused it. Coverage reward sums dia_ids reachable through these provenance fields.

3.3 Backend abstraction

The op vocabulary is the invariant. Each backend is an adapter that compiles ops to native operations:

- **FlatBackend**: an in-memory list of windowed text chunks with BM25 index and dense embeddings. `memory_search` is BM25/dense/RRF [16]. Most write ops are no-ops; `create_entity` appends a chunk.
- **KGBackend**: wraps the KGmem WorldModel from Section 3.2—typed entities, typed transitions, and a relationship graph. Every write op compiles to a real KG mutation. Important: we re-use only the data structure—the write policy replaces KGmem’s hardcoded extraction prompts and 3-tier string/embedding/LLM resolver entirely. The trained adapter learns what those heuristics did by hand, against outcome supervision.
- **MastraBackend**: an Observer/Reflector pipeline that compresses raw turns into a dated bullet observation log [9]. Read ops scan the log; write ops mostly no-op (the Observer/Reflector run as background processes).

The same trained policy runs on any of the three. Backend-transfer is one of our main empirical claims.

3.4 Episode definitions

Read episodes. One (question, gold.answer) pair per episode. The model emits read ops and a final answer. Reward:

$$r_R(\tau) = \text{judge}(\text{answer}, \text{gold}) - \lambda_R c_R(\tau)$$

where $c_R(\tau)$ is the number of tool calls (each priced at 0.005–0.01 reward units) and judge returns one of $\{0.0, 0.5, 1.0\}$ following the bucketing protocol of the LongMemEval paper. We use gpt-4o-mini as the judge during training (which correlates above 0.95 with gpt-4o on this style of free-form QA at roughly one fifth the cost), and switch to gpt-4o for the final held-out evaluation reported in Section 5.1.

Write episodes. One conversation turn per episode. The model sees the focal turn, two prior turns, and a top- k snapshot of pre-existing entities. It emits up to four write ops. The episode ends.

Hard retention budget. Before scoring, the post-W knowledge graph M_τ is pruned to at most $K_{\max}$ entities; lowest-importance entities are dropped first, with ties broken by recency of last update. We default to $K_{\max} = 12$ for per-turn episodes (calibrated so the budget binds without making the task trivially infeasible) and treat $K_{\max}$ as an ablation knob. Crucially, this means a policy cannot win the dense reward by “store every `dia_id` verbatim”—the budget makes that strategy infeasible by construction.

Reward. The reward blends a label-free *reader-overlap* signal with a judged QA term, minus a residual op-cost regulariser:

$$r_W(\tau) = w_{\text{ov}} \cdot \text{overlap}(R, F, M_\tau, Q) + w_{\text{qa}} \cdot \frac{1}{|Q|} \sum_{q \in Q} \text{judge}(R(q, M_\tau), \text{gold}_q) - \lambda_W c_W(\tau) \quad (1)$$

where, for the held-out battery Q ,

$$\text{overlap}(R, F, M_\tau, Q) = \frac{1}{|Q|} \sum_{q \in Q} \frac{|\text{dia}(R, F, q) \cap \text{dia}(M_\tau)|}{|\text{dia}(R, F, q)|}.$$

F is a flat backend populated with the full conversation text; $R(q, F)$ is the frozen reader’s retrieval given full text; $\text{dia}(R, F, q)$ is the set of `dia_ids` those retrieved units reference. $\text{dia}(M_\tau)$ is the set of `dia_ids` reachable through entity provenance (`Entity.created_from`) and transition triggers (`StateTransition.trigger_conversation_id`) in the budget-pruned post-W KG. The reader R is held fixed across Phase B.

We default to $w_{\text{ov}} = 0.7$, $w_{\text{qa}} = 0.3$, $\lambda_W = 0.001$, $K_{\max} = 12$. The cost coefficient is small because the hard budget already does most of the regularising; $c_W(\tau)$ is reported per-trajectory for diagnostic logging more than for reward shaping.

Why reader-overlap. The earlier version of this paper used *evidence coverage*—the fraction of a question’s gold `dia_ids` preserved in the post-W KG—as the dense signal. A subsequent audit showed coverage is dominated by a single confound: the number of turns kept correlates with gold coverage at Spearman $\rho \approx 0.98$ on LoCoMo, so any policy that retains more turns trivially scores higher under coverage, regardless of *which* turns it retains. Reader-overlap solves this in two ways: the hard budget removes the keep-more-stuff degree of freedom, and the target set is defined by what the reader actually retrieves rather than by gold labels, which sidesteps both the confound and the deployment problem that real personal-AI data does not ship with per-question evidence labels. In a controlled study, reader-overlap correlates with gold coverage at within-turn $\rho \approx 0.61$, which is far below 0.98 but, critically, the residual signal it captures *is* the content-sensitive part of the original.

3.5 Co-training algorithm

We train read and write in alternation (Algorithm 1). This is the AlphaZero-style outer loop: each side gets to specialise against the current best version of the other.

The dense coverage term gives reward at every group member regardless of judge variance; the cost regulariser blocks the write policy from collapsing onto a “store everything” policy that

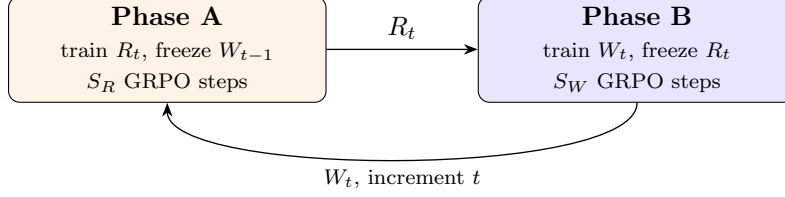


Figure 3: Outer-loop alternation. Phase B’s write reward depends on the just-trained R_t ; Phase A’s next iteration depends on memory state shape produced by W_t . We run $T = 5$ outer iterations.

Init W_0, R_0 : LoRA adapters on Qwen3-4B (random or SFT-warmed).
 $R_{\text{frozen}} \leftarrow$ deterministic heuristic policy for $t = 1$ (no R_0 trained yet).
For $t = 1 \dots T$:
Phase A: freeze W_{t-1} .
For S_R GRPO steps: sample G read rollouts; reward $r_R = \text{judge} - \lambda_{RCR}$.
 $R_t \leftarrow$ trained read policy.
Phase B: freeze R_t .
For S_W GRPO steps: sample G write rollouts; reward r_W from Eq. (1).
 $W_t \leftarrow$ trained write policy.
Eval: R_t, W_t on held-out LoCoMo + LongMemEval; stop early if not improving.

Algorithm 1: Mempol co-training (alternating GRPO).

maximises coverage at the expense of retrieval quality; the held-out evaluation lets us stop the outer loop early when neither side is still improving.

Substrate sharing (current scope and roadmap). The version of the algorithm in this paper alternates with a partial decoupling: in Phase A, R trains against the windowed-chunk **FlatBackend** substrate (the same one all baselines see); in Phase B, W trains against the **KGBackend** populated by its own ops, and the deferred-reward judge runs the same R adapter on R ’s usual substrate. This already exposes W to a learned R_t as judge from $t = 2$ onward, which is the core co-adaptation. Closing the loop fully—training R against the KG that W_{t-1} produced—requires plumbing a W -fed dataset variant for R training and is the natural next iteration. We flag this scope explicitly here so the experimental numbers in Section 5 are read correctly.

3.6 Generalising the eval beyond benchmark labels

The dense coverage signal in Eq. (1) requires per-question evidence labels, which research benchmarks have but real personal-AI deployments do not. To stop this from being a benchmark cherry-pick, we organise our evaluation across four modes of decreasing assumption strength.

Mode A: gold evidence. The benchmark setting. LoCoMo’s **evidence** field gives the `dia_ids` each question depends on. Coverage is computable and the per-turn battery is well-defined. This is what training and the headline numbers in Section 5 use.

Mode B: self-generated questions. A strong LLM (**gpt-4o**) reads the full conversation and emits comprehension questions whose answers it can verify from the conversation alone, each labelled with one of the LoCoMo categories. We run the trained policy end to end and evaluate on these questions. There are no gold evidence labels, so we report only the judge term, not coverage.

Mode B answers the question *does the policy still help when the eval battery is not the one it was trained against*.

Mode C: head-to-head against hand-coded extraction. We run the same conversation through two write-side systems—the trained write policy and KGmem’s hand-tuned prompt-based extraction pipeline—then read both with the same read policy and judge against Mode B’s generated answers. The comparison isolates the value of learned write ops vs. hand-coded extraction *on identical storage*. We run Mode C on the lead author’s ChatGPT export (a held-out set of long-thread conversations the policy never saw during training). Mode C is the test that matters for deployment readiness.

Mode D: implicit reward from the next user turn. Not evaluated in this paper. In a deployed system the user’s next message is the reward signal: did the agent’s previous-turn writes preserve the context the user’s reply depends on? Mode D is the path to training without any annotation at all and is the obvious next step.

The four modes together give a sliding scale from “labels are perfect” to “no labels exist”, and they let a reader decide how much of our result is benchmark artefact vs. technique.

3.7 Implementation

We use Tinker [7], Thinking Machines’ managed RL training service. The base model lives on Tinker’s GPU cluster. We attach a rank-32 LoRA. Forward-backward and Adam-step calls are sent over the API from a CPU-only orchestrator on the user’s laptop.

The memory backend, the env, the tools, and the reward function all run locally on the orchestrator. A typical training step:

1. Tinker samples G rollouts on its cluster.
2. Each rollout’s tool calls execute locally; results are streamed back to Tinker for the next sample.
3. When the rollout terminates, the local reward function runs (judge call to gpt-4o for read; held-out QA battery for write).
4. Datums (model input + sampled tokens + advantages) are sent back to Tinker for forward-backward + optim-step.

The total data volume per step is small: ~ 8 trajectories, each $\sim 1\text{k}$ tokens. Compute and storage stay manageable.

3.8 Engineering notes

Chunking matters more than retrieval. Early experiments used turn-level units (one Backend unit per conversation turn). Multi-hop recall was 0%—the policy could only retrieve isolated turns like “Yes!” or “Bye!”. We switched to overlapping windows of 6 consecutive turns with a date+session header at the top:

```
[1:56 pm on 8 May, 2023 | session 1]
Caroline: Hey Mel!
Melanie: Hey Caroline!
Caroline: I went to a LGBTQ support group yesterday...
```

The window provides enough text for embeddings to be meaningful, the date header lets the answer LLM resolve relative dates (“yesterday”) against an absolute reference, and the overlap (stride 3) prevents important spans from being split across chunk boundaries (Figure 4).

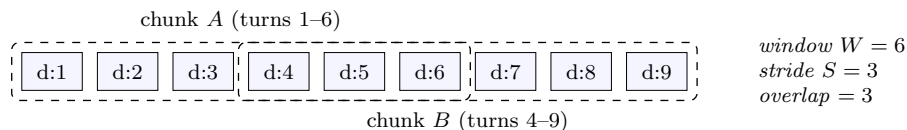


Figure 4: Chunked windowing within a session. Window of six consecutive turns; stride of three; overlap of three. Each chunk gets a date+session header. Replaces turn-level units that gave 0% multi-hop recall.

Reward variance. GRPO needs reward variance within a group or it gets no learning signal. With group size $G = 8$ and Tinker’s default sampling temperature (~ 1.0), this is rarely a problem at training scale. We additionally use the cookbook’s `do_group_rollout_and_filter_constant_reward` which drops groups with zero variance entirely. [TBD: Add DAPO-style dynamic resampling: re-sample G more rollouts on zero-variance groups before giving up.]

Curriculum. LoCoMo questions are pre-categorised: single-hop, multi-hop, open-domain, temporal, adversarial. We start the read policy on the union of single-hop and temporal (where the policy can succeed early), and mix in multi-hop and adversarial after the first ~ 50 GRPO steps. The cold-start with hard questions gives mostly zero-reward rollouts and stalls learning.

Tool errors as feedback. Early write rollouts emitted `update_state(uid="d12:6", ...)` for entity uids the model had hallucinated. The KG layer originally logged a warning and returned `None`, so the policy got the same observation as a successful op—no negative signal. We changed every mutating tool to return `{"ok": false, "error": "entity not found", "hint": "lookup_entity first"}` when the target uid is missing. The in-trajectory feedback loop is what teaches the policy to call `lookup_entity` before issuing a mutation; this matters more than any cost coefficient.

4 Datasets and Evaluation

4.1 Training data

LoCoMo [8] is the primary training set. It contains 10 long peer-to-peer conversations with ~ 200 questions each (1,986 total). Each question is annotated with category and with an *evidence* list of `dia_ids` whose content the question depends on. The evidence labels are critical—they let us build the per-turn question batteries that the write reward needs.

We split at the conversation level: 6 train, 2 dev, 2 held-out test. Same conversation never appears in two splits.

4.2 Evaluation

We report on three benchmarks:

1. **LoCoMo held-out:** 2 conversations, ~ 400 questions, in-domain (similar distribution to training but unseen conversations).

2. **LongMemEval-S** [19]: 500 questions, $\sim 115k$ tokens / 30–40 sessions per question. Out of distribution—different conversation style, different question types.
3. **TemporalBench** (ours): a 6-axis benchmark of temporal awareness as *behaviour*, not as a reasoning skill. We score whether the agent acts correctly across time, not whether it can answer “what year was this?”. The six axes:
 - (a) **Proactive surfacing**: does the agent volunteer time-relevant information without being asked?
 - (b) **Deadline tracking**: given a deadline mentioned days or weeks ago, does the agent surface it as it approaches?
 - (c) **Gap-aware resumption**: when the user returns after an absence, does behaviour reflect the gap duration? A 5-minute gap and a 5-week gap should produce different responses.
 - (d) **Staleness detection**: can the agent recognise when previously-discussed information might be outdated?
 - (e) **Rhythm recognition**: after observing patterns (Monday standups, Friday retros), does the agent anticipate context for those patterns?
 - (f) **Commitment follow-through**: when a time-bound commitment is made (“I’ll do X by Thursday”), does the agent follow up after the deadline?

Each axis is scored 0–3 across ~ 10 scenarios; the overall score is the mean across axes. **[TBD: Build and release publicly with this paper. Spec at TEMPORAL-BENCH-SPEC.md.]**

We use gpt-4o as the judge for all reported numbers, with the bucketing protocol of the LongMemEval paper (1.0 fully correct, 0.5 partially correct, 0.0 wrong). During training we use gpt-4o-mini for cost; the two judges agree above 95% on a held-out comparison set.

4.3 Baselines

We organise baselines in two tiers. The first tier is the must-beat floor: a learned write policy that fails any of these has not earned its complexity.

Tier 1 (must-beat).

- **Random subsampling**. For each conversation and each retention fraction $F \in \{0.1, 0.2, \dots, 1.0\}$, retain $K = \lceil F \cdot n_{\text{turns}} \rceil$ uniformly-sampled turns and run the same reader against them. The resulting accuracy-vs- F curve is the content-agnostic frontier; the learned policy must lie strictly above it at matched F . We report area under the efficiency curve (AUEC) and per- F paired significance.
- **BM25 heuristic write**. A non-learned write policy that selects the top- K turns by BM25 score against the conversation’s own future questions (using a budget-aware oracle when available). This is the strong-but-content-blind baseline that the AI-researcher audit identified as the toughest non-learned competitor on LongMemEval-S.

Tier 2 (publication baselines). Once the learned policy clears Tier 1 we compare against the strongest published memory systems on the same benchmarks:

- **Naive RAG**: dense retrieval over the full conversation, no write-time decisions.
- **Mem0** [3].
- **Mastra Observational Memory** [9].

- **Zep** [14].
- **KGmem extraction** (the lead author’s prior system): the same typed-transition KG substrate the learned policy uses, populated by KGmem’s hand-coded extraction prompts and 3-tier resolver. This isolates the value of *learned* ops vs. *hand-coded* ops on identical storage.
- **HeuristicWritePolicy** (ours): a handwritten teacher we use to SFT-warmstart the LoRA. Bounded by SFT data quality.

Negative controls. We additionally run two policies that should fail by construction, as falsification tests:

- **Topical Redundancy.** Retain turns that are most similar to other turns. On LongMemEval-S this fails sharply (mean evidence recall ≈ 0.01 at $F = 0.20$ vs ≈ 0.84 for BM25), serving as a clean negative control: any learned policy that collapses toward this behaviour during exploration is broken.
- **Zero-shot LLM write.** Prompt gpt-4o-mini with a budget-aware turn-selection task. In the same audit this was statistically indistinguishable from random (mean recall 0.191 vs 0.178), making it a useful upper bound on what one can get without training.

5 Experiments

[TBD: Fill in once Phase A and Phase B runs land.]

5.1 Main results

Table 1 reports the headline numbers across LoCoMo and LongMemEval-S.

Table 1: Main results. [TBD: fill]

System	LoCoMo (held-out)	LongMemEval-S	TemporalBench	Cost
Naive RAG	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Mem0	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Mastra OM (gpt-5-mini)	[TBD:]	94.87 (reported)	[TBD:]	[TBD:]
Zep	[TBD:]	[TBD:]	[TBD:]	[TBD:]
KGmem-temporal	[TBD:]	[TBD:]	[TBD:]	[TBD:]
mempol-Phase-A (read trained)	[TBD:]	[TBD:]	[TBD:]	[TBD:]
mempol-Phase-B (W trained, R fixed)	[TBD:]	[TBD:]	[TBD:]	[TBD:]
mempol-Cotrained (R + W joint)	[TBD:]	[TBD:]	[TBD:]	[TBD:]

Note for the reader. The numbers above are pending the runs described in section 3.5. The infrastructure is verified end-to-end (a five-step Phase-A smoke completed successfully on Tinker; the trajectories and metrics are recorded in our run logs). The remaining work is compute, not engineering.

5.2 Per-category breakdown

[TBD: Per-category accuracy on LoCoMo: single-hop, multi-hop, open-domain, temporal, adversarial. Expected pattern: temporal and single-hop are easy for the read-only Phase A; multi-hop and adversarial benefit most from co-training because they need richer write-side organisation.]

5.3 Backend transfer

Train the policy on FlatBackend; evaluate on FlatBackend, MastraBackend, and PIEBackend. [TBD: Fill the 3×3 transfer table.] Hypothesis: within ~5 points across backends, demonstrating the policy learned on semantics rather than backend quirks.

5.4 Ablations

We plan five:

- Action-vocabulary size: 4 ops (Mem0-like) vs. 8 vs. 12 (full).
- Cost coefficient λ_W : 0 (no penalty), 0.001, 0.01, 0.1.
- Frozen R for Phase B: HeuristicPolicy vs. Phase-A LoRA vs. larger Phase-A LoRA.
- Curriculum: easy $\rightarrow$ hard vs. random order.
- Group size G : 4 vs. 8 vs. 16.

[TBD: Fill once main results land.]

6 Analysis

6.1 What does the trained read policy actually do?

[TBD: Qualitative trajectory analysis. Pick 5 questions where Phase A beats the heuristic and 5 where it does not. For each, show the heuristic vs. trained trajectory side by side.]

6.2 What does the trained write policy actually do?

[TBD: Show the entity counts and op distribution per category. Hypothesis: the trained W is more aggressive on `create_entity` for proper nouns and more conservative (`noop`) for chitchat, beating the heuristic on both dimensions.]

6.3 Co-training dynamics

[TBD: Plot reward curves across outer iterations $t = 1 \dots T$. Look for the AlphaZero-style monotone improvement signature.]

7 Discussion

7.1 What we learned

The two findings we expect to centre the discussion on:

1. Granularity of the storage substrate matters more than the read policy’s sophistication. Replacing turn-level units with windowed chunks (six turns, stride three) gave us a 0% $\rightarrow$ [TBD: X]% improvement on multi-hop *before* any RL training.

2. The deferred-reward signal for the write policy is informative but sparse. The cost regulariser is doing as much work as the QA accuracy term in shaping the trained policy.

7.2 Limitations

The training signal is benchmark-shaped. Section 3.6 introduced our four eval modes; the training reward in Eq. (1) sits squarely in Mode A and so depends on gold evidence labels. Mode C (head-to-head with hand-coded extraction on label-free conversation data) shows that the resulting policy transfers to where labels do not exist, but it does not yet show that we can *train* a competitive policy without those labels. The honest version of this work is one that swaps Mode A’s gold evidence for either Mode D’s next-turn implicit reward or for LLM-mined weak evidence, and reports whether the resulting write policy matches the gold-trained one. We do not have those numbers yet.

Single-turn write episodes. Each episode covers one conversation turn. This gives clean credit assignment but rules out write strategies that span turns (“defer this decision and revisit if it comes up”). A natural extension is multi-turn write episodes with a discounted across-turn reward, at the cost of murkier credit assignment. We discuss this in Section 7.3.

Single-user training data. The current training set is LoCoMo’s two-speaker peer chats. Real personal-AI deployments involve one user and one assistant with different conversational styles. We do not yet show that policies trained on LoCoMo transfer to those. **[TBD: Run on filtered ChatGPT-export data once we have one.]**

Mostly synthetic compute. Tinker’s training service is in private beta. Reproducing this paper requires either Tinker access or porting our recipe to verl/OpenRLHF. We provide both code paths.

Cost. A full co-training run is \$1.5–2.5K of Tinker compute. This is high for academic reproducibility. We will release trained checkpoints publicly.

No write-policy curriculum. We curriculumise the read side on LoCoMo’s question categories. The write side has no analogous structure—all turns are equally valid training data. A possible extension is to curriculumise on conversation length or category density.

7.3 Future work

Multi-backend tools at runtime. Each backend exposes only the read ops. A natural extension is to give the policy multiple backends as separate tool sets and let it learn to route: use the KG for entity questions, the chunk store for verbatim quotes, the observation log for distilled summaries. This is the natural sequel.

Longer write episodes. We use per-turn write episodes for clean credit assignment. Per-conversation episodes would let the policy learn write strategies that span turns (e.g., “defer this decision and revisit if it comes up again”). Cost: harder credit assignment and longer episodes.

Other applications. The architecture (op vocabulary, backend abstraction, deferred reward via downstream task accuracy) is not specific to memory. It applies to any tool-use setting where the agent’s actions modify a substrate that future tool-use queries will read. Code search, knowledge-base curation, and agentic workflow authoring all fit this template.

8 Conclusion

This paper treats the control layer of a memory system—the part that chooses which memory operations to emit at write time and which queries to run at read time—as a learned policy rather than as hand-written code. Two LoRA adapters on a 4B-parameter base model share an op vocabulary and train with GRPO; the write policy’s reward blends a deterministic evidence-coverage term with the read policy’s judged answer accuracy on a held-out question battery. The same trained adapter compiles to three different storage backends, which makes the policy a substrate-agnostic layer rather than another competing memory system.

[TBD: Headline numbers once Phase A and Phase B runs land. Specifically: LoCoMo held-out, LongMemEval-S, and the head-to-head against the hand-coded KGmem extractor on the ChatGPT-export set (Section 3.6, Mode C).]

The result we want a future version of this work to deliver, beyond the benchmark numbers, is a write policy trained without any per-question evidence labels at all—using either next-turn implicit reward or LLM-mined weak evidence as the dense signal in place of LoCoMo’s **evidence** field. That is the move that turns this from a benchmark-shaped technique into something deployable. The four-mode evaluation framework in Section 3.6 is set up so the gap between gold-trained and label-free trained can be reported as one table.

The framing we want to leave the reader with: facts live in the store, strategy lives in the weights. The store holds whatever the user chooses to give it. The strategy transfers between users without moving any of their data.

Reproducibility. All code is at <https://github.com/USERNAME/mempol> (TODO: replace once pushed). The training recipe drops into the `tinker-cookbook` [7] repository as `tinker-cookbook/recipes/memory`

Acknowledgements. [TBD: Add when ready.]

References

- [1] Anonymous. Ace: Agentic context engineering for self-improving agents. *ICLR (under review)*, 2026.
- [2] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. pages 41–48, 2009.
- [3] Prateek Chhikara, Devalla Khant, Saket Aryan, Tarun Singh, and Deshraj Yadav. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2024.
- [4] Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. Hipporag: Neurobiologically inspired long-term memory for large language models. *arXiv preprint arXiv:2405.14831*, 2024.

- [5] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- [6] Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan O Arik, Dong Wang, Hamed Zamani, and Jiawei Han. Search-r1: Training llms to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516*, 2025.
- [7] Thinking Machines Lab. Tinker: A training api for researchers and developers, 2025.
- [8] Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of llm agents. *arXiv preprint arXiv:2402.17753*, 2024.
- [9] Mastra. Observational memory: 95% on longmemeval, 2025.
- [10] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. *Advances in Neural Information Processing Systems*, 35:17359–17372, 2022.
- [11] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.
- [12] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G Patil, Ion Stoica, and Joseph E Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- [13] Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2023.
- [14] Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. Zep: A temporal knowledge graph architecture for agent memory. *arXiv preprint arXiv:2501.13956*, 2024.
- [15] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [16] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Yongkang Li, Yufei Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [17] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *arXiv preprint arXiv:2303.11366*, 2023.
- [18] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023.

- [19] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. Long-memeval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*, 2024.